

EECE 306 Lab 2 notebook. Coordinate Systems, Points and Vector Components

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Fill in every TODO, run `publish('Lab02_notebook.m')` from the toolbox root, print the HTML to PDF, three to five pages.

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Setup

```
clear; close all;  
rng(306);
```

Point roundtrips, 1000 random points

```
r = randn(1000,3);  
esph = max(em.vec.mag(em.coord.sph2c(em.coord.c2sph(r)) - r));  
ecyl = max(em.vec.mag(em.coord.cyl2c(em.coord.c2cyl(r)) - r));  
fprintf('spherical roundtrip max error = %.3e\n', esph);  
fprintf('cylindrical roundtrip max error = %.3e\n', ecyl);
```

```
spherical roundtrip max error = 2.812e-15  
cylindrical roundtrip max error = 2.979e-15
```

Poles and the phi convention

```
disp('c2sph([0 0 1]) ='); disp(em.coord.c2sph([0 0 1]))  
disp('c2sph([0 0 -1]) ='); disp(em.coord.c2sph([0 0 -1]))  
q = [ 1 1 0; -1 1 0; -1 -1 0; 1 -1 0 ];  
p = em.coord.c2cyl(q);  
fprintf('phi in the four quadrants = %.4f %.4f %.4f %.4f rad\n', p(:,2));  
fprintf('all inside [0, 2*pi) = %d\n', all(p(:,2) >= 0 & p(:,2) < 2*pi));
```

```
c2sph([0 0 1]) =  
    1    0    0
```

```
c2sph([0 0 -1]) =  
    1.0000    3.1416    0
```

```
phi in the four quadrants = 0.7854 2.3562 3.9270 5.4978 rad
all inside [0, 2*pi) = 1
```

Vector component roundtrips and the invariant

```
A = randn(1000,3);
As = em.coord.vecC2Sph(A, r);
ev = max(em.vec.mag(em.coord.vecSph2C(As, r) - A));
em_ = max(abs(em.vec.mag(As) - em.vec.mag(A)));
fprintf('vector roundtrip (spherical pair) max error = %.3e\n', ev);
fprintf('magnitude change under conversion max      = %.3e\n', em_);
Ac = em.coord.vecC2Cyl(A, r);
ev2 = max(em.vec.mag(em.coord.vecCyl2C(Ac, r) - A));
fprintf('vector roundtrip (cylindrical pair) max error = %.3e\n', ev2);
```

```
vector roundtrip (spherical pair) max error = 1.088e-15
magnitude change under conversion max      = 8.882e-16
vector roundtrip (cylindrical pair) max error = 6.280e-16
```

The two position demonstration

The same Cartesian vector $A = 2x + 3y$, expressed in spherical components at two different positions.

```
A0 = [2 3 0];
disp('at (0, 5, 0), (Ar, Atheta, Aphi) ='); disp(em.coord.vecC2Sph(A0, [0 5 0]))
disp('at (4, 0, 0), (Ar, Atheta, Aphi) ='); disp(em.coord.vecC2Sph(A0, [4 0 0]))
% TODO two to four sentences. The vector did not change. Explain exactly
% what did change, and why a position argument is therefore mandatory for
% component conversion and meaningless for point conversion.
```

```
at (0, 5, 0), (Ar, Atheta, Aphi) =
    3.0000    0.0000   -2.0000
```

```
at (4, 0, 0), (Ar, Atheta, Aphi) =
    2.0000    0.0000    3.0000
```

Interpretation

TODO what class of bug does the roundtrip test catch, and what class does it miss. Name the invariant that catches what the roundtrip misses, and say why it is reference free.

Problems encountered

TODO honest account, or NONE.

Full test suite

```
%{
runTests
EECE 306 toolbox test suite
```

```
c =
```

```
2.9979e+08
```

```
eta =
```

```
376.7303
```

```
test_lab01 checks complete
```

```
PASS test_lab01      0.06 s
```

```
PASS test_lab02      0.19 s
```

```
2 passed, 0 failed, 2 total
```

```
All tests passing.
```

```
%}
```